

Group Project (30 marks)

Warning – Copyright Notice

*All materials related to this assignment are the intellectual property of Griffith University and are protected by copyright law. **Students are strictly prohibited from uploading, sharing, or publishing any part of this assignment to any public online platforms.** Griffith University reserves the right to investigate and take appropriate action against any student found to have breached this policy.*

Background

The electricity company **XPower** wants to develop a simple **Household Tariff Analysis software (desktop, android/apple app, or website)** that helps customers:

- Upload their household electricity consumption data.
- Calculate electricity bills under different tariff schemes.
- Compare the costs across tariff schemes and identify savings.
- Visualize bills and usage trends.
- Receive cost-saving suggestions.
- Operate the system via a friendly user interface (UI) (desktop or web).

Project Constraints (given by XPower):

- **Project Duration:** 5 weeks
- **Budget:** \$10,000

Your team has accepted this project. You are required to develop a working software prototype with a friendly UI and apply project management knowledge to manage the entire process.

Requirements provided by XPower:

1. Data Import
 - Support CSV/Excel files containing daily or hourly electricity usage.
2. Tariff Models
 - Flat rate tariff (single unit price). (**Details as below**)
 - Time-of-use tariff (peak / off-peak / shoulder). (**Details as below**)
 - Tiered tariff (different unit prices for consumption thresholds). (**Details as below**)
3. Bill Calculation & Comparison
 - Given a duration, generate bills with detailed breakdown (fixed fee, energy fee).
 - Given a duration, compare multiple tariff schemes, highlighting cost differences and savings.
4. Visualization & Reporting for a selected period
 - Line charts: electricity usage trend.
 - Bar or pie charts: bill breakdown.
5. User Interface (UI)
 - A simple, user-friendly interface (desktop or web).

Users should upload files and run analyses without needing technical skills.

Tariff Models

a) Flat Rate Tariff

- **Definition:** A single fixed price per unit of electricity (kWh), regardless of when it is consumed or how much is consumed.
- **Requirements:**
 - User can input the flat rate value (e.g., \$0.25/kWh).
 - The system multiplies the total kWh consumed in the billing period by the flat rate.
 - A **fixed monthly supply fee** (if specified) should also be included.
- **Example:**
 - Consumption: 300 kWh/month
 - Flat rate: \$0.25/kWh
 - Fixed fee: \$10
 - Total bill = $(300 \times 0.25) + 10 = \85

b) Time-of-Use (TOU) Tariff

- **Definition:** Different unit prices depending on the time of day when electricity is consumed. Typically divided into **Peak**, **Off-Peak**, and **Shoulder** periods.
- **Requirements:**
 - User must define the time periods and their rates (e.g., Peak 6pm–10pm \$0.40/kWh, Off-Peak 10pm–7am \$0.15/kWh, Shoulder all other times \$0.25/kWh).
 - The system should parse each usage record's timestamp and apply the correct rate.
 - The bill should show a breakdown by tariff category (Peak/Off-Peak/Shoulder).
- **Example:**
 - Usage:
 - Peak: $100 \text{ kWh} \times \$0.40 = \40
 - Shoulder: $120 \text{ kWh} \times \$0.25 = \30
 - Off-Peak: $80 \text{ kWh} \times \$0.15 = \12
 - Fixed fee: \$10
 - Total bill = $40 + 30 + 12 + 10 = \$92$

c) Tiered (Block) Tariff

- **Definition:** Different unit prices apply depending on the total consumption volume, with thresholds (tiers) set for cumulative usage.
- **Requirements:**
 - User defines tier thresholds and rates (e.g., Tier 1: first 100 kWh \$0.20, Tier 2: 101–300 kWh \$0.30, Tier 3: above 300 kWh \$0.40).
 - The system calculates cost progressively across tiers.
 - The bill should show how much consumption fell into each tier.
- **Example:**
 - Consumption: 350 kWh/month
 - Rates:

- First 100 kWh × \$0.20 = \$20
- Next 200 kWh × \$0.30 = \$60
- Remaining 50 kWh × \$0.40 = \$20
- Fixed fee: \$10
- Total bill = 20 + 60 + 20 + 10 = \$110

Group Work Policy

1. Contribution and Assessment

- Students who contribute equally to the project will receive the same mark.
- Students whose contribution is less than their peers will receive a proportionally lower mark.
- Students who make **no contribution** to the project will receive a mark of **zero**.

2. Submission

- Each group is required to submit **one final version** of the project.
- Resubmission is allowed.
- It is recommended that the **group leader** upload the submission on behalf of the group.

3. Collaboration

- Groups are encouraged to collaborate by sharing and co-editing a common file (e.g., *Group_project.docx*) to ensure that all members are able to contribute effectively.

4. Conflict Resolution

- In the event of conflicts within the group (e.g., disagreements, lack of contribution, miscommunication), members should first attempt to resolve the issue through internal discussion.
- If the issue cannot be resolved internally, the matter should be referred to the course convenors for mediation.
- The convenors reserve the right to adjust individual marks if required to ensure fairness.

The following table must be included on the first page of your main submission document.

Group 000

Student No.	Full Name	Contribution (sum to 100%)	Signature
s111111	Full Name	33.3% / Equal	
s222222	Full Name	33.3% / Equal	
s333333	Full Name	33.3% / Equal	
Explanation:			

Your team should apply appreciate project management knowledge throughout the project lifecycle to complete this software development project. You are required to carry out the following procedures, following the standards and guidelines introduced in lectures and workshops.

1. Initiation

- Write a **Project Charter**.
- Identify stakeholders at least three stakeholders.

2. Planning

- **Work Breakdown Structure (WBS)**: Develop a hierarchical WBS with at least three levels. Provide a WBS Dictionary for a minimum of **four** selected work packages.
- **Project Schedule**: Define **all** activities, estimate their durations, determine activity sequencing, and produce a complete **Gantt chart** to illustrate the schedule. (Please include the screenshot in your main submission document.)
- **Risk Register**: Identify and document at least **four** project risks.
- **Quality Metrics**: Define at least **four** quality metrics that will be used to evaluate the project's deliverables and overall success.

3. Execution

- **Implement the software**
 - **Use Case Diagram**: Create a system-level Use Case Diagram. (Please include the screenshot in your main submission document.)
 - **Use Case Descriptions**: Provide at least **four** detailed Use Case Descriptions.
 - **High-Level Architectural Design**: Present an architectural design that shows how system components interact. The architecture should cover all identified requirements and features. (Please include the screenshot in your main submission document.)
 - **Coding**: **You are not required to implement the entire system or all requirements**. Instead, implement at least **Three simple** functions in Python (testing will also be conducted in Python). These Three functions will be the basis for testing. Among these:
 - At least one function must include parameters.
 - At least one function must return a value.
 - At least one function must use an if–else structure.
- **Conduct testing**
 - **Unit Testing**
 - For each implemented function, design both positive and negative test cases to thoroughly validate its behaviour.
 - Must achieve a **100% pass rate**.
 - Please include the screenshot of the test results in your main submission document.

4. Closing

- Each team member must individually prepare a **Lessons Learned Report (250–500 words)**, covering, but not limited to: project experience, reflections on project management, teamwork and collaboration, etc.

Files required for submission (Only files in DOCX, PDF, or ZIP format are permitted):

1. Group_Project.docx or pdf (This file **must not** be included in the zip file)
2. All **source code** files (.py), including both function code and testing code, must be compressed into a single ZIP file.

Please refer to the submission page for the Marking Rubric.